

Part 2: Billing UI Analysis

Question A – UI Mock-up Review

Security

1. **Missing CVV/CVC (Security Code) field**

Without it, many payment gateways decline transactions or flag them as higher fraud risk.

2. **Raw card data in plain text fields**

The card number appears in a standard input on the merchant page, not inside a payment provider's secure iframe. For a modern global SaaS company, this typically increases breach and compliance risk.

3. **No visible security/trust indicators**

There is no lock icon, "Secured by [provider]" badge, or indication that payment data is encrypted/tokenized, which affects trust.

4. **Redundant manual card type selection**

Brand can be auto-detected from the card number; forcing users to choose VISA/Mastercard/etc. Mismatch between selected type and entered number can cause failed payments.

Usability

1. **Inconsistent required-field marking**

Expiration **Year** is marked required (*), but **Month** is not—both are needed for a valid expiry date.

2. **Not suitable for a global SaaS product**

There is no **Country** field. The state dropdown placeholder says "Select a state," which is very US-centric despite the label "global SaaS company". International users cannot complete billing correctly.

3. **Poor input formatting UX**

Labels instruct users not to use dashes or spaces (card number, postal code). The form should accept common formats and normalize input automatically.

4. **Payment amount lacks context**

"30.00" is shown without currency (USD, EUR, etc.), billing period, or description of what is being charged.

5. **Awkward Middle Initial (MI) layout**

The MI field is cramped between first and last name, making the form look wierd.

Compliance / Business

- No authorization text (e.g. "I authorize a charge of \$30.00").
 - No link to terms, refund policy, or subscription details before payment.
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Question B – Sample Test Cases

TC-001: Valid submission with all required fields

Priority	P1 - Critical
Type	Positive
Preconditions	User on billing screen; Payment Amount = 30.00
Test Data	VISA · 4111111111111111 · Exp 12/2028 · John Smith · 123 Main St · Boston · MA · 02108

Step	Action	Expected Result
1	Fill all required fields with test data above	All values accepted
2	Click Continue	Form submits; user reaches confirmation/next step with no errors

TC-002: Required-field validation on empty submit

Priority	P1 - Critical
Type	Negative
Preconditions	User on billing screen; all fields empty

Step	Action	Expected Result
1	Click Continue without entering data	Form does not submit
2	Verify required fields	Errors shown for: Card Type, Card Number, Exp Month, Exp Year, First Name, Last Name, Address Line 1, City, State, Postal Code
3	Verify optional fields (MI, Address Line 2)	No errors when empty
4	Verify focus	Focus moves to first invalid field

TC-003: Invalid card number rejected

Priority	P2 - High
Type	Negative
Preconditions	User on billing screen; all other fields valid (TC-001 data)

Step	Action	Expected Result
1	Enter 4111-1111-1111-1111	Dashes stripped or clear error shown
2	Enter 411111 (too short)	Validation error; form blocked
3	Enter 41111111111111112 (invalid Luhn)	Validation error; form blocked
4	Click Continue with invalid number	No submission; no payment API call

Question C – Product Solution for the Most Severe Bug

Most severe bug: No CVV/CVC field. The form can't complete a normal card payment and skips a key fraud check.

Fix: Add a required **Security code (CVV)** field next to expiration (3 digits for Visa/Mastercard, 4 for Amex). Validate on client and server; send it only over HTTPS to your payment provider; **never store or log it** (PCI requirement). Use tokenization so your app keeps a payment token, not raw card data.

More modern option: Embed a **third-party payment iframe** instead of building raw card fields yourself. The iframe keeps card data in the provider's PCI-compliant environment, reduces your compliance scope, and gives you built-in validation, CVV handling, and fraud tools.

Done when: Submit is blocked without CVV; valid CVV allows payment; invalid/missing CVV fails; CVV never appears in your DB or logs.